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Exercise list: L05 — Introduction to causal inference (Classes 5+6): textbook exercises

Instructions. These are the *Practice* items of Classes 5+6 in `Map/class-plan.md`. Every question opens with a source line. Book statements (Wooldridge, Greenberg) are reproduced verbatim from the book PDF. *Mostly Harmless Econometrics* has no end-of-chapter exercises: for each of its worked examples the table is reproduced verbatim and the question is original. Computer exercises are solved in Python (`statsmodels`, `linearmodels`); `Listas/L05-book-exercises-check.py` reproduces and asserts every number in the solutions. Show your steps.

Block 1 — MHE worked examples *(MHE Ch. 2–6 • Questions 1–6)*

Question 1. The STAR experiment: balance, selection bias and covariates (MHE Tables 2.2.1–2.2.2). *Source:* Angrist and Pischke (2008), *Mostly Harmless Econometrics: An Empiricist's Companion*, §2.2, Tables 2.2.1 and 2.2.2, printed pp. 14–15. The source has no exercise; the statement is original. The table numbers are reproduced verbatim from the PDF (table notes shortened).

Table 2.2.1: Comparison of treatment and control characteristics in the Tennessee STAR experiment

		Students who entered STAR in kindergarten			
	Variable	Small	Regular	Regular/Aide	Joint P-value
1.	Free lunch	.47	.48	.50	.09
2.	White/Asian	.68	.67	.66	.26
3.	Age in 1985	5.44	5.43	5.42	.32
4.	Attrition rate	.49	.52	.53	.02
5.	Class size in kindergarten	15.10	22.40	22.80	.00
6.	Percentile score in kindergarten	54.70	48.90	50.00	.00

Notes (excerpt): "Adapted from Krueger (1999), Table 1. The table shows means of variables by treatment status. The P-value in the last column is for the F-test of equality of variable means across all three groups."

Table 2.2.2: Experimental estimates of the effect of class-size assignment on test scores

Explanatory variable	(1)	(2)	(3)	(4)
Small class	4.82	5.37	5.36	5.37
	(2.19)	(1.26)	(1.21)	(1.19)
Regular/aide class	.12	.29	.53	.31
	(2.23)	(1.13)	(1.09)	(1.07)
White/Asian (1 = yes)	–	–	8.35	8.44
			(1.35)	(1.36)
Girl (1 = yes)	–	–	4.48	4.39
			(.63)	(.63)
Free lunch (1 = yes)	–	–	-13.15	-13.07
			(.77)	(.77)
White teacher	–	–	–	-.57
				(2.10)
Teacher experience	–	–	–	.26
				(.10)
Master's degree	–	–	–	-0.51
				(1.06)
School fixed effects	No	Yes	Yes	Yes
R^2	.01	.25	.31	.31

Note (excerpt): "The dependent variable is the Stanford Achievement Test percentile score. Robust standard errors that allow for correlated residuals within classes are shown in parentheses. The sample size is 5681."

(a) What null hypothesis does the "Joint P-value" column test? Using rows 1–3 of Table 2.2.1, did randomization balance the groups at the 5% level? Why should rows 4–6 not be read as balance checks?

(b) Write the observed difference in mean scores between small and regular classes as the effect on the treated plus selection bias, and explain why random assignment makes the coefficient on *Small class* an estimate of the average causal effect.

(c) Using column (1) of Table 2.2.2, compute the t statistic and a 95% confidence interval for the small-class effect; repeat with column (4). Is the regular/aide effect in column (1) significant?

(d) From column (2) to column (3) the small-class coefficient barely moves while its standard error falls from 1.26 to 1.21. Explain both facts, and show that the fall in the standard error is about what the rise in R^2 from .25 to .31 predicts.

(e) The text says the effect is about $.2\sigma$, where σ is the standard deviation of the kindergarten percentile score. What σ does column (4) imply? Is it plausible for a percentile scale?

Solution:

Question 2. Returns to schooling and omitted-variables bias (MHE Table 3.2.1).
Source: Angrist and Pischke (2008), *Mostly Harmless Econometrics*, §3.2.2, Table 3.2.1, printed p. 62. The source has no exercise; the statement is original. Table reproduced verbatim from the PDF.

Table 3.2.1: Estimates of the returns to education for men in the NLSY

	(1)	(2)	(3)	(4)	(5)
Controls:	None	Age dummies	Col. (2) and additional controls*	Col. (3) and AFQT score	Col. (4), with occupation dummies
	0.132	0.131	0.114	0.087	0.066
	(0.007)	(0.007)	(0.007)	(0.009)	(0.010)

Notes: "Data are from the National Longitudinal Survey of Youth (1979 cohort, 2002 survey). The table reports the coefficient on years of schooling in a regression of log wages on years of schooling and the indicated controls. Standard errors are shown in parentheses. The sample is restricted to men and weighted by NLSY sampling weights. The sample size is 2434. *Additional controls are mother's and father's years of schooling and dummy variables for race and Census region."

(a) Write the long regression of log wages on schooling, the column (3) controls and AFQT, and the auxiliary regression of AFQT on schooling and the same controls. Derive the omitted-variables-bias (OVB) formula that links the schooling coefficients in columns (3) and (4).

(b) Compute the change in the schooling coefficient attributable to AFQT, and to the family-background controls (column (2) to column (3)). What do the signs say about the correlations involved?

(c) Suppose the auxiliary regression of AFQT (in standard-deviation units) on schooling, with the column (3) controls, had slope 0.15 (a hypothetical value, not in the book). What effect of one standard deviation of AFQT on log wages does the OVB formula then imply?

(d) Column (5) is half of column (1). Why is column (5) *not* the most credible estimate of the return to schooling?

(e) Give a 95% confidence interval for the return to schooling in column (4).

Solution:

Question 3. Wald and 2SLS estimates: quarter of birth and the draft lottery (MHE Tables 4.1.1–4.1.3).

Source: Angrist and Pischke (2008), *Mostly Harmless Econometrics*, §4.1, Tables 4.1.1, 4.1.2 and 4.1.3, printed pp. 92, 96 and 97. The source has no exercise; the statement is original. Table numbers reproduced verbatim from the PDF (notes shortened).

Table 4.1.1: 2SLS estimates of the economic returns to schooling

	OLS		2SLS					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Years of education	0.075	0.072	0.103	0.112	0.106	0.108	0.089	0.061
	(0.0004)	(0.0004)	(0.024)	(0.021)	(0.026)	(0.019)	(0.016)	(0.031)
Covariates:								
Age (in quarters)								✓
Age (in quarters) squared								✓
9 year of birth dummies		✓			✓	✓	✓	✓
50 state of birth dummies		✓			✓	✓	✓	✓
Instruments:			dummy for QOB=1	dummy for QOB=1 or QOB=2	dummy for QOB=1	full set of QOB dummies	full set of QOB dummies int. with year of birth dummies	full set of QOB dummies int. with year of birth dummies

Notes (excerpt): "The table reports OLS and 2SLS estimates of the returns to schooling using the the Angrist and Krueger (1991) 1980 Census sample. [. . .] The sample size is 329,509. Robust standard errors are reported in parentheses."

Table 4.1.2: Wald estimates of the returns to schooling using quarter of birth instruments

	(1) Born in the 1st or 2nd quarter of year	(2) Born in the 3rd or 4th quarter of year	(3) Difference (std. error) (1)-(2)
ln (weekly wage)	5.8916	5.9051	-0.01349
			(0.00337)
Years of education	12.6881	12.8394	-0.1514
			(0.0162)
Wald estimate of return to education			0.0891
			(0.0210)
OLS estimate of return to education			0.0703
			(0.0005)

Notes (excerpt): "Adapted from a re-analysis of Angrist and Krueger (1991) by Angrist and Imbens (1995). [. . .] The sample size is 329,509."

Table 4.1.3: Wald estimates of the effects of military service on the earnings of white men born in 1950

	Earnings		Veteran Status		Wald Estimate
Earnings year	Mean (1)	Eligibility Effect (2)	Mean (3)	Eligibility Effect (4)	of Veteran Effect (5)
1981	16,461	-435.8	0.267	0.159	-2,741
		(210.5)		(0.040)	(1,324)
1971	3,338	-325.9			-2050
		(46.6)			(293)
1969	2,299	-2.0			
		(34.5)			

Notes (excerpt): "Adapted from Angrist (1990), Tables 2 and 3. Standard errors are shown in parentheses. [...] Figures are in nominal dollars. [...] There are about 13,500 individuals in the sample."

(a) Derive the Wald estimator for a binary instrument and state the assumptions under which it identifies a causal effect.

(b) Table 4.1.2: compute the Wald estimate from column (3) and check it against 0.0891. Approximate its standard error treating the first stage as known, and compare with 0.0210.

(c) Table 4.1.3: compute the Wald estimates for 1981 and 1971 from columns (2) and (4), their approximate standard errors, and each as a share of mean earnings. What is the 1969 row for?

(d) Table 4.1.1: is the 2SLS estimate in column (3) significantly different from the OLS estimate in column (1)? Use a Hausman-type statistic with $V(\hat{\rho}_{IV} - \hat{\rho}_{OLS}) = V(\hat{\rho}_{IV}) - V(\hat{\rho}_{OLS})$. Why is the standard error in column (8) larger than in columns (6)–(7), even though (8) uses as many instruments?

Solution:

Question 4. Card and Krueger: difference-in-differences by hand (MHE Table 5.2.1).
Source: Angrist and Pischke (2008), *Mostly Harmless Econometrics*, §5.2, Table 5.2.1, printed p. 171. The source has no exercise; the statement is original. Table reproduced verbatim from the PDF.

Table 5.2.1: Average employment per store before and after the New Jersey minimum wage increase

	Variable	PA (i)	NJ (ii)	Difference, NJ-PA (iii)
1.	FTE employment before,	23.33	20.44	-2.89
	all available observations	(1.35)	(0.51)	(1.44)
2.	FTE employment after,	21.17	21.03	-0.14
	all available observations	(0.94)	(0.52)	(1.07)
3.	Change in mean FTE	-2.16	0.59	2.76
	employment	(1.25)	(0.54)	(1.36)

Notes (excerpt): "Adapted from Card and Krueger (1994), Table 3. The table reports average full-time equivalent (FTE) employment at restaurants in Pennsylvania and New Jersey before and after a minimum wage increase in New Jersey. [...] Standard errors are reported in parentheses"

(a) Compute the difference-in-differences estimate twice: as the difference of the two states' changes, and as the change in the NJ–PA gap. Reconcile with 2.76.

(b) Write the regression $Y_{ist} = \alpha + \gamma NJ_s + \lambda d_t + \delta(NJ_s \cdot d_t) + \varepsilon_{ist}$ and give the value of each coefficient implied by the table. State the identifying assumption.

(c) Compute the t statistic and a 95% interval for the DiD estimate, and show that its standard error is what independent samples in the two states imply.

(d) Is the sign of the estimate what a competitive labor-demand model predicts?

Solution:

Question 5. Incumbency and the sharp regression-discontinuity design (MHE Figure 6.1.2).
Source: Angrist and Pischke (2008), *Mostly Harmless Econometrics*, §6.1, Figure 6.1.2 and text, printed pp. 194–195. The source has no exercise; the statement is original. Caption and numbers from the text reproduced verbatim; the two fitted values at the cutoff are read off the figure (not printed in the book).

Figure 6.1.2 (caption, verbatim): "Probability of winning an election by past and future vote share (from Lee, 2008). (a) Candidate's probability of winning election $t + 1$, by margin of victory in election t: local averages and parametric fit. (b) Candidate's accumulated number of past election victories, by margin of victory in election t: local averages and parametric fit." The text states that the winner is determined by $D_i = 1(x_i \geq .0)$, where x_i is the vote share margin of victory; that the dots are "the average win rate in non-overlapping windows of share margins that are .005 wide"; that the lines are fitted from "a Logit model for the probability of winning as a function of the cutoff indicator $D_i = 1(x_i \geq 0)$, a 4th-order polynomial in x_i, and interactions between the polynomial terms and D_i"; and that "incumbency appears to raise party re-election probabilities by about 40 percentage points." In panel (a) the fitted curve reaches about 0.14 just left of zero and about 0.58 just right of zero; in panel (b) the two fitted curves nearly meet at zero (both close to one past victory).

(a) Write the sharp RD model with a polynomial control and define the estimand as a jump in a conditional expectation. State the identifying assumption.

(b) Estimate the incumbency effect from panel (a) and compare it with "about 40 percentage points".

(c) What does panel (b) test, and what does it show?

(d) Give two reasons why an RD estimate could be spurious here, and a check for each.

Solution:

Question 6. Maimonides' rule and fuzzy RD as IV (MHE Table 6.2.1).
Source: Angrist and Pischke (2008), *Mostly Harmless Econometrics*, §6.2, Table 6.2.1 and text, printed pp. 199–202. The source has no exercise; the statement is original. Table and the rule reproduced verbatim from the PDF.

The text defines the predicted class size from Maimonides' rule (class size capped at 40) as

$$m_{sc} = \frac{e_s}{\text{int}\left[\frac{(e_s-1)}{40}\right] + 1},$$

"where int(x) is the integer part of a real number, x", e_s is enrollment in the grade, and estimates by 2SLS "an equation like $y_{isc} = \alpha_0 + \alpha_1 d_s + \beta_1 e_s + \beta_2 e_s^2 + \cdots + \beta_p e_s^p + \rho n_{sc} + \eta_{isc}$" (6.2.6), with n_{sc} class size, d_s the proportion of disadvantaged students, and m_{sc} as instrument.

Table 6.2.1: OLS and fuzzy RD estimates of the effects of class size on fifth grade math scores

	OLS			2SLS				
				Full sample		Discontinuity samples		
						+/- 5		+/- 3
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mean score		67.3		67.3		67.0		67.0
(s.d.)		(9.6)		(9.6)		(10.2)		(10.6)
Regressors								
Class size	.322	.076	.019	-.230	-.261	-.185	-.443	-.270
	(.039)	(.036)	(.044)	(.092)	(.113)	(.151)	(.236)	(.281)
Percent disadvantaged		-.340	-.332	-.350	-.350	-.459	-.435	
		(.018)	(.018)	(.019)	(.019)	(.049)	(.049)	
Enrollment			.017	.041	.062		.079	
			(.009)	(.012)	(.037		(.036)	
Enrollment squared/100					-.010			
					(.016)			
Segment 1 (enrollment 36-45)								-12.6
								(3.80)
Segment 2 (enrollment 76-85)								-2.89
								(2.41)
Root MSE	9.36	8.32	8.30	8.40	8.42	8.79	9.10	10.2
R-squared	.048	.249	.252					
N		2,018			2,018		471	302

Notes: "Adapted from Angrist and Lavy (1999). The table reports estimates of equation (6.2.6) in the text using class averages. Standard errors, reported in parentheses, are corrected for within-school correlation." (The missing parenthesis in "(.037" is in the original.)

(a) Compute m_{sc} for enrollments $e_s = 40, 41, 80, 81, 120, 121$. Why does the rule generate a *fuzzy* rather than a sharp RD?

(b) Write the first and second stages of the 2SLS model. Which variables play the roles of the instrument, the running variable and the treatment?

(c) Explain the sign and the movement of the OLS class-size coefficient across columns (1)–(3).

(d) Using column (4) and the text's round slope of $-.25$, how much would a 7-pupil cut in class size raise math scores, in points and in standard deviations of the class-average score (9.6)? Give the t statistics in columns (4) and (8).

(e) Compare the ratio of the standard errors in columns (8) and (5) with $\sqrt{2018/302}$. What does the comparison say?

Solution:

Block 2 — Instrumental variables with the CARD data *(Wooldridge Ch. 15 • Questions 7–8)*

Question 7. Wooldridge, Computer Exercise 15.C3.

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 15, Computer Exercise C3, printed pp. 492–493, reproduced verbatim.

C3 Use the data in CARD for this exercise.

(i) The equation we estimated in Example 15.4 can be written as

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \cdots + u,$$

where the other explanatory variables are listed in Table 15.1. In order for IV to be consistent, the IV for *educ*, *nearc*4, must be uncorrelated with u. Could *nearc*4 be correlated with things in the error term, such as unobserved ability? Explain.

(ii) For a subsample of the men in the data set, an IQ score is available. Regress *IQ* on *nearc*4 to check whether average IQ scores vary by whether the man grew up near a four-year college. What do you conclude?

(iii) Now, regress *IQ* on *nearc*4, *smsa*66, and the 1966 regional dummy variables *reg*662, ..., *reg*669. Are *IQ* and *nearc*4 related after the geographic dummy variables have been partialled out? Reconcile this with your findings from part (ii).

(iv) From parts (ii) and (iii), what do you conclude about the importance of controlling for *smsa*66 and the 1966 regional dummies in the log(*wage*) equation?

Solution:

Question 8. Wooldridge, Computer Exercise 15.C5.

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 15, Computer Exercise C5, printed p. 493, reproduced verbatim.

C5 Use the data in CARD for this exercise.

(i) In Table 15.1, the difference between the IV and OLS estimates of the return to education is economically important. Obtain the reduced form residuals, $\hat{v}_2$, from the reduced form regression *educ* on *nearc*4, *exper*, *exper*^{2}, *black*, *smsa*, *south*, *smsa*66, *reg*662, ..., *reg*669—see Table 15.1. Use these to test whether *educ* is exogenous; that is, determine if the difference between OLS and IV is statistically significant.

(ii) Estimate the equation by 2SLS, adding *nearc*2 as an instrument. Does the coefficient on *educ* change much?

(iii) Test the single overidentifying restriction from part (ii).

Solution:

Block 3 — Difference-in-differences *(Wooldridge Ch. 13 • Questions 9–11)*

Question 9. Wooldridge, Problem 13.2.

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 13, Problem 2, printed p. 425, reproduced verbatim. Equation (13.9), which the problem refers to, is reproduced from p. 410 for reference.

2 Using the data in KIELMC, the following equations were estimated using the years 1978 and 1981:

$$\widehat{\log(price)} = \underset{(.26)}{11.49} - \underset{(.058)}{.547}\, nearinc + \underset{(.080)}{.394}\, y81 \cdot nearinc$$
$$n = 321,\ R^2 = .220$$

and

$$\widehat{\log(price)} = \underset{(.27)}{11.18} + \underset{(.044)}{.563}\, y81 - \underset{(.067)}{.403}\, y81 \cdot nearinc$$
$$n = 321,\ R^2 = .337.$$

Compare the estimates on the interaction term $y81 \cdot nearinc$ with those from equation (13.9). Why are the estimates so different?

For reference, equation (13.9), p. 410:

$$\widehat{\log(price)} = \underset{(.31)}{11.29} + \underset{(.045)}{.457}\, y81 - \underset{(.055)}{.340}\, nearinc - \underset{(.083)}{.063}\, y81 \cdot nearinc, \qquad n = 321,\ R^2 = .409. \quad (13.9)$$

Solution:

Question 10. Wooldridge, Problem 13.6.

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 13, Problem 6, printed p. 426, reproduced verbatim.

6 In 1985, neither Florida nor Georgia had laws banning open alcohol containers in vehicle passenger compartments. By 1990, Florida had passed such a law, but Georgia had not.

(i) Suppose you can collect random samples of the driving-age population in both states, for 1985 and 1990. Let *arrest* be a binary variable equal to unity if a person was arrested for drunk driving during the year. Without controlling for any other factors, write down a linear probability model that allows you to test whether the open container law reduced the probability of being arrested for drunk driving. Which coefficient in your model measures the effect of the law?

(ii) Why might you want to control for other factors in the model? What might some of these factors be?

(iii) Now, suppose that you can only collect data for 1985 and 1990 at the county level for the two states. The dependent variable would be the fraction of licensed drivers arrested for drunk driving during the year. How does this data structure differ from the individual-level data described in part (i)? What econometric method would you use?

Solution:

Question 11. Wooldridge, Computer Exercise 13.C3.

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 13, Computer Exercise C3, printed p. 427, reproduced verbatim.

C3 Use the data in KIELMC for this exercise.

(i) The variable *dist* is the distance from each home to the incinerator site, in feet. Consider the model

$$\log(price) = \beta_0 + \delta_0 y81 + \beta_1 \log(dist) + \delta_1 y81 \cdot \log(dist) + u.$$

If building the incinerator reduces the value of homes closer to the site, what is the sign of δ_1? What does it mean if $\beta_1 > 0$?

(ii) Estimate the model from part (i) and report the results in the usual form. Interpret the coefficient on $y81 \cdot \log(dist)$. What do you conclude?

(iii) Add *age*, age^2, *rooms*, *baths*, $\log(intst)$, $\log(land)$, and $\log(area)$ to the equation. Now, what do you conclude about the effect of the incinerator on housing values?

(iv) Why is the coefficient on $\log(dist)$ positive and statistically significant in part (ii) but not in part (iii)? What does this say about the controls used in part (iii)?

Solution:

Block 4 — Bayesian treatment models *(Greenberg Ch. 11 • Questions 12–15)*

Context for Questions 12–15 (Greenberg 2007, §11.1, printed pp. 169–172; the model and the results the exercises refer to, reproduced verbatim, selected passages). The response is related to the covariates and the treatment assignment through

$$y_{0i} = x_i'\beta_0 + u_{0i}, \qquad y_{1i} = x_i'\beta_1 + u_{1i}, \tag{11.1}$$

"where x_i is a K_1 vector of covariates, a 0 subscript indicates assignment to the control group, and 1 indicates assignment to the treatment group." Assignment is modeled by

$$s_i^* = x_i'\gamma_1 + z_i'\gamma_2 + v_i, \qquad s_i = \begin{cases} 1, & \text{if } s_i^* > 0, \\ 0, & \text{if } s_i^* \le 0, \end{cases} \tag{11.2}$$

$$y_i = (1 - s_i)y_{0i} + s_i y_{1i}, \tag{11.3}$$

"where z_i is a $K_2 \times 1$ vector of instrumental variables."

$$\text{Cov}(u_{0i}, v_i) \equiv \Sigma_0 = \begin{pmatrix} \sigma_{00} & \sigma_{0v} \\ \sigma_{0v} & 1 \end{pmatrix}, \qquad \text{Cov}(u_{1i}, v_i) \equiv \Sigma_1 = \begin{pmatrix} \sigma_{11} & \sigma_{1v} \\ \sigma_{1v} & 1 \end{pmatrix},$$

"where $\sigma_{vv} = 1$ because of the probit specification." "Let $\beta = (\beta_0', \beta_1', \gamma_1', \gamma_2')'$, $\gamma = (\gamma_1', \gamma_2')'$, $X_{ji} = \begin{pmatrix} (1-j)x_i' & jx_i' & 0 & 0 \\ 0 & 0 & x_i' & z_i' \end{pmatrix}$, [...] $N_0 = \{i : s_i = 0\}$, $N_1 = \{i : s_i = 1\}$, and n_0 and n_1, respectively, are the number of observations in N_0 and N_1. We assume a Gaussian prior distribution for $\beta \sim N_p(b_0, B_0)$, where $p = 3K_1 + K_2$. [...] we reparameterize the variances in Σ_j as $\omega_{jj} = \sigma_{jj} - \sigma_{jv}^2$ [...]. We define $\psi_j = (\omega_{jj}, \sigma_{jv})$ and specify the prior distributions $\pi(\psi_j) = \pi(\omega_{jj})\pi(\sigma_{jv} \mid \omega_{jj}) = \text{IG}(\nu_{j0}/2, d_{j0}/2)N(m_{j0}, \omega_{jj}M_{j0})$."

$$y_{ji} \mid s_i^*, \beta_j, \gamma, \psi_j \sim N(x_i'\beta_j + \sigma_{jv}e_i, \omega_{jj}), \tag{11.4}$$

"where $e_i = s_i^* - x_i'\gamma_1 - z_i'\gamma_2$. Let Y_j, X_j, Z_j, S_j^*, and E_j, respectively, contain the values of y_{ji}, x_i, z_i, s_i^*, and e_i for $i \in N_j$. Then"

$$Y_j \mid S_j^*, \beta_j, \gamma, \psi_j \sim N_{n_j}(X_j\beta_j + \sigma_{jv}E_j,\ \omega_{jj}I_{n_j}). \tag{11.5}$$

$$E(Y_j \mid S_j^*, \beta_j, \gamma, \omega_{jj}) = X_j\beta_j + m_{j0}E_j, \tag{11.6}$$

$$\text{Var}(Y_j \mid S_j^*, \gamma, \omega_{jj}) = \omega_{jj}(I_{n_j} + E_jM_{j0}E_j'). \tag{11.7}$$

$$f(y_{ji}, s_i = j \mid x_i, z_i, \beta, \psi_j) = N(y_{ji} \mid x_i'\beta_j, \sigma_{jj})\,\Phi\left((2j-1)\frac{\mu_{ji}}{\phi_j}\right). \tag{11.8}$$

Algorithm 11.1: Gibbs algorithm for treatment model (Chib, in press-a)

11.1 Sample ψ_j conditioned on (y, s^*, X, Z, β)

(a) Drawing ω_{jj} marginalized over σ_{jv} from $\text{IG}\left(\frac{\nu_{j0}+n_j}{2}, \frac{d_{j0}+d_j}{2}\right)$, where $d_j = (Y_j - X_j\beta_j - m_{j0}E_j)'(I_{n_j} + E_jM_{j0}E_j')^{-1}(Y_j - X_j\beta_j - m_{j0}E_j)$,

(b) Drawing σ_{jv} conditioned on ω_{jj} from $N(c_j, \omega_{jj}C_j)$, where $c_j = C_j[M_{j0}^{-1}m_{j0} + E_j'(Y_j - X_j\beta_j)]$, $C_j = (M_{j0}^{-1} + E_j'E_j)^{-1}$.

11.2 Sample β conditioned on (y, s^*, ψ_0, ψ_1) from $N_p(\hat{\beta}, B)$, where $\hat{\beta} = B(B_0^{-1}b_0 + \sum_{i\in N_0} X_{0i}'\Sigma_0^{-1}y_i^* + \sum_{i\in N_1} X_{1i}'\Sigma_1^{-1}y_i^*)$, $B = (B_0^{-1} + \sum_{i\in N_0} X_{0i}'\Sigma_0^{-1}X_{0i} + \sum_{i\in N_1} X_{1i}'\Sigma_1^{-1}X_{1i})^{-1}$, and $y_i^* = (y_i, s_i^*)'$.

11.3 Sample s^* conditioned on $(y, X, Z, \beta, \psi_0, \psi_1)$ by drawing s_i^* from $\text{TN}_{(-\infty,0)}(\mu_{0i}, \phi_0^2)$ if $s_i = 0$, $\text{TN}_{(0,\infty)}(\mu_{1i}, \phi_1^2)$ if $s_i = 1$, where $\mu_{ji} = x_i'\gamma_1 + z_i'\gamma_2 + \sigma_{jv}\sigma_{jj}^{-1}(y_{ji} - x_i'\beta_j)$, $\phi_j^2 = 1 - \sigma_{jv}^2\sigma_{jj}^{-1}$.

Common notation for the solutions. $j \in \{0,1\}$: untreated/treated; Φ: standard normal c.d.f.; $\mathrm{IG}(\alpha,\beta)$ has density proportional to $\omega^{-(\alpha+1)}e^{-\beta/\omega}$ $(\omega>0)$; $\mathrm{TN}_{(a,b)}$: normal truncated to (a,b). The bivariate-normal conditioning rule (Greenberg, Appendix A, eq. A.9) is used throughout: if $(a,b)'$ is bivariate normal,

$$a \mid b \sim N\!\left(\mu_a + \frac{\sigma_{ab}}{\sigma_{bb}}(b-\mu_b),\ \sigma_{aa} - \frac{\sigma_{ab}^2}{\sigma_{bb}}\right). \tag{A.9}$$

Question 12. Greenberg, Exercise 11.1.

Source: Greenberg (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 11, Exercise 11.1, printed p. 180, reproduced verbatim.

11.1 Verify (11.4).

Solution:

Question 13. Greenberg, Exercise 11.2.

Source: Greenberg (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 11, Exercise 11.2, printed p. 180, reproduced verbatim.

11.2 Verify (11.6) and (11.7).

Solution:

Question 14. Greenberg, Exercise 11.3.

Source: Greenberg (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 11, Exercise 11.3, printed p. 181, reproduced verbatim.

11.3 Verify (11.8).

Solution:

Question 15. Greenberg, Exercise 11.4.

Source: Greenberg (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 11, Exercise 11.4, printed p. 181, reproduced verbatim.

11.4 Verify the conditional posterior distributions used in Algorithm 11.1.

Solution: